
PROFESSIONAL SUMMARY

Policy researcher with a background in data science, public policy, consulting, and economic research. Experienced in using econometric and statistical methods to analyze large-scale survey, market, and administrative data, with research spanning labor markets, social policy, digital infrastructure, and economic development. Skilled in R and Python, with strengths in causal inference, survey analysis, time-series methods, data visualization, and translating complex analysis into clear insights for policymakers and senior decision-makers.

EDUCATION

COLUMBIA UNIVERSITY, School of International and Public Affairs (SIPA)

Master of Public Administration, Data Science for Policy (*Urban & Social Policy Minors*)

May 2026

Relevant Coursework: Causal Inference & Program Evaluation, Applied Econometrics, Time Series Analysis, Applied Machine Learning, Microeconomics, Macroeconomics, Urban Economics, Comparative Social Welfare Policy

KING FAHD UNIVERSITY OF PETROLEUM & MINERALS, Engineering College

Bachelor of Science, Electrical Engineering (*Signal Processing & Artificial Intelligence Minors*)

September 2021

RESEARCH EXPERIENCE

INTERNATIONAL LABOUR ORGANIZATION (ILO)

Graduate Researcher

November 2025 - Present

- Analyzed nationally representative survey data covering nearly 70,000 individuals to examine labor-market vulnerability, socioeconomic conditions, demographic disparities, and the drivers of forced labour.
- Designed and implemented survey-weighted statistical analyses in R, incorporating complex survey design and sampling weights to produce representative estimates and evaluate variation across demographic and socioeconomic groups.
- Applied econometric methods, including logistic regression, propensity score matching, and marginal-effects estimation, to identify relationships between labor vulnerability and factors including age, education, migration, household characteristics, indebtedness, economic shocks, and social protection.
- Conducted geographic analysis across six regions to identify spatial patterns in labor vulnerability and differences in demographic and economic profiles across locations.
- Developed a data-driven typology of affected populations using Gower distance, PAM clustering, and multidimensional scaling, translating complex individual-level data into interpretable population profiles for policy targeting.
- Produced charts, heatmaps, geographic comparisons, statistical tables, and other analytical visualizations to communicate patterns across population groups and regions.
- Maintained documented and reproducible R workflows through GitHub for data cleaning, transformation, statistical analysis, diagnostics, and visualization using tidyverse, survey, ggplot2, MatchIt, and cobalt.
- Contributed to research reports, policy briefs, and technical documentation, translating statistical results into clear findings and policy implications for technical and nontechnical audiences.

PROFESSIONAL EXPERIENCE

COMMUNICATION, SPACE & TECHNOLOGY COMMISSION (CST)

Technology Strategist

April 2023 - August 2024

- Designed a quantitative performance framework for 32 cloud service providers using primary data collected through provider surveys and interviews, combining market and operational indicators into a performance index that informed approximately \$30M in national investment policy and supported market benchmarking.
- Analyzed data from more than 1,200 organizations across nine industries to identify market trends, performance gaps, demand patterns, and differences across industry segments, translating findings into recommendations for investment and regulatory policy.
- Built Tableau dashboards and automated performance-monitoring tools that enabled leadership to track market indicators and provider performance in real time, reducing reporting and decision-making time by approximately 60%.
- Conducted comparative analysis of provider and industry performance to identify market disparities, benchmark outcomes, and assess changes in the competitive landscape.
- Collaborated across policy, investment, engineering, and regulatory teams to translate quantitative analysis into five reforms to the national cloud regulatory framework.
- Prepared analytical reports, policy briefs, data visualizations, and executive presentations for senior government stakeholders.
- Developed data-driven infographics and public-facing analytical materials that translated technical and market information into accessible insights for broader audiences.

DELOITTE

Business Analyst

April 2022 - April 2023

- Conducted large-scale analytical assessments for a public sector client, integrating quantitative data, stakeholder inputs, and operational information to identify investment needs, performance gaps, and opportunities for organizational improvement.
- Analyzed services and infrastructure across 230 systems for a financial-sector client, identifying operational gaps, regulatory risks, and resource requirements to support long-term infrastructure and investment planning.
- Developed predictive models to estimate workload and resource demand, improving planning accuracy and supporting data-driven allocation of infrastructure resources.
- Conducted a current-state assessment across 11 government sub-agencies, analyzing service delivery, systems, and infrastructure to identify inefficiencies and inform the integration of 66 systems, contributing to a 20% reduction in operating costs.
- Produced client-ready analyses and presented quantitative findings, risks, and strategic recommendations to senior leadership, translating complex technical evidence into implementation priorities.

STC

Access & Transport Strategy Developer

December 2021 - April 2022

- Conducted geospatial and quantitative analysis of regional infrastructure and service-access data to identify geographic disparities and inform investment planning.
- Assessed differences in connectivity and infrastructure access across regions, supporting strategies to expand service coverage and improve digital inclusion.
- Produced analytical reports and visualizations translating regional and infrastructure data into strategic recommendations for senior decision-makers.

ARAMCO

Control and Network Engineer

Ras Tanura, Saudi Arabia

September 2021 - December 2021

- Supported the migration and modernization of mission-critical industrial control systems underpinning large-scale energy export operations, including SCADA, PLC, and DCS architectures.
- Coordinated with cross-functional engineering and safety teams to align control systems with reliability, cybersecurity, and operational governance standards.

LEADERSHIP & ACTIVITIES

Program Assistant – Data Science for Policy Concentration at SIPA

Sep 2025 - May 2026

- Coordinated academic programming, workshops, and events bridging data science and public policy.
- Collaborated with faculty, administrators, and students to improve program operations and enhance student engagement.
- Supported curriculum development and resources to strengthen applied data science training for policy students.
- Led hands-on workshops mentoring students in Python and R programming, data manipulation, and applied analytics.

CERTIFICATIONS

- **Advanced Regulations** - University of Oxford, Oriel College
Completed an intensive program on advanced regulatory frameworks, including antitrust, competition law, and governance mechanisms. Developed expertise in navigating complex regulatory environments, enhancing strategic decision-making, and applying theoretical knowledge to practical regulatory challenges.
- **Digital Regulations** - International Telecommunication Union (ITU)
Completed a professional program on digital regulation, covering regulatory governance, competition, consumer protection, data privacy, spectrum management, emerging technologies, two-sided platforms, and digital inclusion. Gained practical skills in applying regulatory frameworks, developing evidence-based policies, and navigating the complexities of the digital economy.

SKILLS

- **Statistical & Econometric Methods:** OLS and generalized linear regression, logistic regression, fixed effects, difference-in-differences, instrumental variables (2SLS), regression discontinuity, propensity score matching, time series modeling and ARIMA-based forecasting, and GARCH and VAR models.
- **Machine Learning:** Decision Trees, Random Forest, LASSO/Ridge Regularization, k-Nearest Neighbor, PAM, and K-Modes Clustering
- **Programming & Tools:** R (tidyverse, ggplot2, MatchIt, plm, ivreg, knitr, survey), Python (pandas, sklearn, statsmodels), Stata, SQL, GitHub, MS Office Suite (advanced), Tableau, Canva, Figma.
- **Research & Communication:** Policy briefs, client-facing analytical reports, technical documentation, codebook development, data visualization, and executive presentations.